

Assessment and Quantification of Ice Mass Change in Greenland from GRACE and GRACE-FO Observations*

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Abstract—The accelerated melting of Greenland’s glaciers, along with a decline in wintertime accumulation rates, represents worrying symptoms of climate change. While this phenomenon has long been observed, accurately quantifying the associated ice mass loss has remained a challenge. Since the launch of the GRACE mission in 2002, high-precision satellite gravimetry has enabled researchers to monitor changes in Earth’s gravity field and by extension variations in ice mass. This paper aims to quantify and compare ice mass loss across distinct regions of Greenland between 2004 and 2017. Using GRACE and GRACE-FO Level-2 data, spherical harmonic expansions are applied to derive gravity anomalies. A low-pass Gaussian filter is then applied to reduce noise, followed by least-squares regression to analyze annual and seasonal trends. The analysis reveals that Greenland lost a total of 1,580 km³ of ice over the study period, corresponding to an average equivalent water height loss of 0.869 m. The South-West region experienced the greatest loss, with 376.42 km³ of ice melted between 2004 and 2017.

Nomenclature

Symbols

a	m	Radius of the Earth
$\mathbf{B}$		Block matrix of basis functions
C_{lm}		Cosine spherical harmonics coefficient
ΔC_{lm}		Change in C_{lm}
g_r	m/s ²	Radial gravitational acceleration
k_l		Love load number
P_{lm}		Associated Legendre function
$\bar{P}_{lm}$		Normalized associated Legendre function
R_e	m	Earth’s mean radius
Res		Residual
r	m	Radial distance to center of body
r_G	m	Gaussian smoothing radius
r_i		Residual
SH_{val}		GRACE spherical harmonic value
S_{lm}		Sine spherical harmonics coefficient

ΔS_{lm}		Change in S_{lm}
$\mathbf{T}$		Basis function vector
U	m ² /s ²	Gravitational potential
$\mathbf{W}$		Weighting matrix
$\mathbf{X}$		Regression coefficient vector
$\mathbf{Y}$		Observation vector
y_i		Observed spherical harmonics coefficient
$\hat{y}_i$		Modeled spherical harmonics coefficient
α	m/s ²	Acceleration
$\Delta\sigma$	kg/m ³	Change in surface density
θ	rad	Colatitude
ϕ	rad	Smoothing angle
φ	rad	Latitude
λ	rad	Colongitude
μ	m ³ /s ²	Standard gravitational parameter
ρ_{avg}	kg/m ³	Average Earth density
ρ_{water}	kg/m ³	Density of water
σ_i		Standard deviations of observations

Subscripts

0	Reference point
i	Index of observation in a series
m	Order
l	Degree

Abbreviations

EWH	Equivalent Water Height
GRACE	Gravity Recovery and Climate Experiment
GRACE-FO	GRACE Follow-On
LSQR	Least Squares Regression

I. Introduction

Ice caps are influenced by global warming and thus serve as key indicators of climate change [1]. Their mass balance reflects shifts in temperature and precipitation patterns. Understanding the temporal variations in their growth and decline is thus essential for predicting future climate trends and assessing the impacts of global warming. One

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key aspect of this dynamic is the winter growth rate of said ice caps. Changes in this rate over time may indicate more general climate shifts, with implications for water resources, sea-level contributions, and regional climate patterns. Despite being able to observe this effect, it was not possible to quantify this until recently.

Since 2002, more precise estimates can be made with the use of GRACE and GRACE-FO data [2]. GRACE (-FO) consists of a pair of satellites, in an orbit of around 500 km around Earth with an inter-satellite distance of approximately 220 km [3]. These satellites precisely measure inter-satellite distance up to micron-level accuracy [3]. These precise measurements of the inter-satellite distance can be related to gravity field deviations. The change of mass can be related to the measured local gravitational field changes with the help of gravitational models. Those models predict the gravity field around the Earth, which by common misconception is believed to be a perfect sphere. In reality, due to the collection of mass in certain regions of the Earth and the resulting change in local strength of the gravitational field [4], the spherical assumption of the gravitational field is not accurate.

Hence, the data received from GRACE (-FO) helps build a more complete and accurate gravitational model [2]. The deviations from the spherical approximation can be modeled with the help of spherical harmonics and related to the surface of the global mean sea level, or geoid. GRACE (-FO) level 2 data already provides two parameters describing the geoid, one of which takes the surface of the Earth into account, while the second one focuses on the solid Earth. Those parameters are summed up to end up with the change of mass of the Earth's surface expressed in equivalent water height (EWH) [5]. However, the level 2 data incorporates errors when relating the changes in the gravity field to the change in equivalent water height. Therefore, filters need to be applied to the data in order to correct for those errors and give a proper reading of the data. [5].

There has been a lot of research on mass loss of single glaciers using GRACE(-FO) data [1, 6, 7], however, there has been little research in comparing the different regions of Greenland with the same model. This makes it difficult to fairly compare the mass loss between the different regions within Greenland. For this reason, this paper has the aim to quantify and compare ice mass loss across distinct regions of Greenland between 2004 and 2017. This will be achieved using the above described GRACE(-FO) data.

II. Method

A. Data and Spherical Harmonics

To analyze the mass changes, a dataset which consists of monthly estimates of Earth's geopotential field, derived from GRACE and GRACE-FO missions, is used. Specifically, the release 06.1 (RL06.1) Level-2 spherical harmonic solutions provided by the NASA Jet Propulsion Laboratory is used [8]. This data represents time-variable gravity field solutions in the form of fully normalized spherical harmonic coefficients, computed from inter-satellite ranges, measured by the GRACE-FO Microwave Instrument. Each solution corresponds to an average over approximately one month, which correlates to the mass distribution of the Earth. The RL06.1 is an updated release of the RL06 data, only differing in the accelerometer data used for the GRACE-FO. Improving the quality of the gravitational field estimates whilst maintaining the compatibility and format of the initial RL06 release [8].

The gravitational acceleration over Earth's surface was computed using the fully normalized spherical harmonic coefficients provided in the aforementioned dataset [8]. These coefficients were applied in Equation 1, which was evaluated over a global grid defined in colongitude λ and colatitude θ . In this computation, the associated Legendre polynomials P_l^m were generated using the `scipy.special` library [9]. However, due to numerical instability at higher degrees and orders, the evaluation was limited to $l, m \leq 90$. These associated Legendre polynomials also need to be normalized. Following the definitions provided by Yuan et al. [10] and Holmes et al. [11], the Legendre polynomials were normalized using Equation 2, where $k = 1$ for $m = 0$ and $k = 2$ for $m > 0$.

$$U(r, \lambda, \theta) = \frac{\mu}{R_e} \cdot \sum_{l=0}^{\infty} \sum_{m=0}^l \left(\frac{R_e}{r} \right)^{l+1} \cdot (\bar{C}_l^m \cos(m\lambda) + \bar{S}_l^m \sin(m\lambda)) \cdot \bar{P}_l^m(\cos(\theta)) \quad (1)$$

$$\bar{P}_l^m = P_l^m \cdot \sqrt{k(2l+1) \frac{(l-m)!}{(l+m)!}} \quad (2)$$

The gravitational potential obtained from the spherical harmonic expansion was then used to compute the gravitational acceleration by taking its radial derivative, as shown in Equation 3. This yielded the distribution of gravitational acceleration over the Earth's surface, visualized in Figure 1.

$$g_r(r, \lambda, \theta) = \frac{\mu}{R_e} \cdot \sum_{l=0}^{\infty} \sum_{m=0}^l \frac{R_e^{l+1}}{r^{l+2}} \cdot (l+1) \cdot (\bar{C}_l^m \cos(m\lambda) + \bar{S}_l^m \sin(m\lambda)) \cdot \bar{P}_l^m(\cos(\theta)) \quad (3)$$

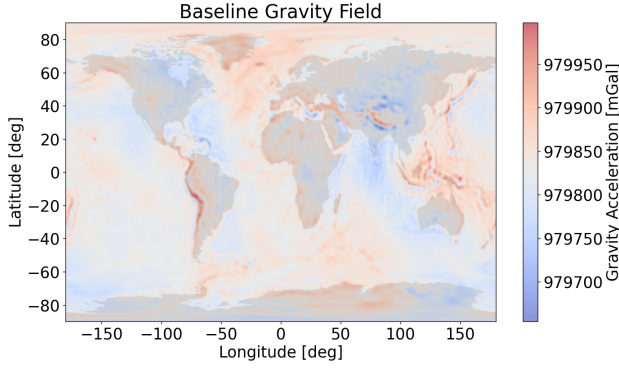


Fig. 1. Gravitational acceleration over Earth's surface, plotted at 360×180 resolution in milligals (mGal), based on data from January 2004 to December 2017.

B. Filtering

When generating difference maps, by subtracting monthly gravity fields, to observe temporal variations, characteristic North–South striping appears, as shown in Figure 2. This striping does not have a geophysical origin, but results from correlated errors in the spherical harmonic coefficients, instrument noise, and low signal-to-noise ratios [12]. These artifacts, predominantly oriented in the North–South direction, are introduced during data processing and obscure the true geophysical signals.

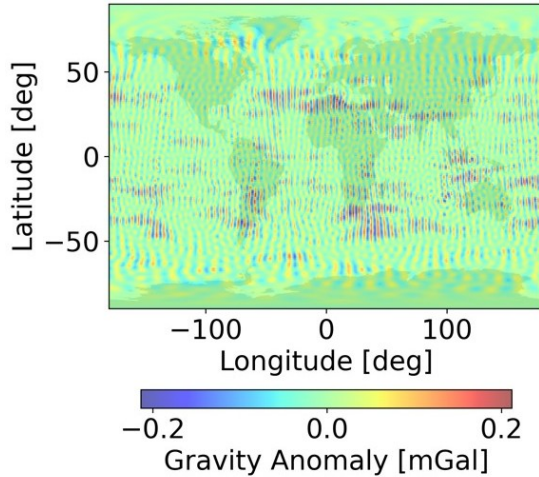


Fig. 2. Unfiltered change in gravitational acceleration over Earth's surface, shown at 360×180 resolution in milligals (mGal), based on data from April–May 2002 and May–June 2017.

There are numerous types of filters, ranging from isotropic and anisotropic filters to empirical orthogonal function techniques. In this study, two filters are selected: one to remove striping and another to smooth the data. Specifically, an error-correlated filter is applied first, followed by an isotropic Gaussian filter.

The correlated error filter functions by grouping all spherical harmonic coefficients of the same order m . For each group, a polynomial of a certain degree is fitted to approximate the correlated noise. This polynomial, representing the striping pattern, is subtracted from the original spherical harmonic coefficients [12]. This procedure is typically not applied to coefficients of order $m = 0$, or those with $m > 30$, where striping is less prominent [12].

The main goal of this filter is to significantly reduce the North–South striping. As an anisotropic filter, it applies directional smoothing, specifically targeting said North–South artifacts [13]. While several anisotropic filters are available, this correlated error filter is selected due to its simplicity and effectiveness.

After the striping is reduced, an isotropic Gaussian filter is used to smooth out the remaining irregularities. Isotropic filters smooth data uniformly in all directions and are, therefore, directionally independent [13]. The Gaussian filter emphasizes large-scale, low-degree coefficients by assigning them higher weights, while reducing the influence of high-degree (noisy) coefficients through lower weights [14]. The general weighting function is shown in Equation 4 [15].

$$W_l = \exp\left(-\frac{l(l+1)\phi^2}{2}\right) \quad (4)$$

In Equation 4, W_l denotes the weight applied to degree l . The parameter ϕ , defined in Equation 5, is the smoothing angle expressed in radians and serves as a scaling factor, where r_G is the Gaussian smoothing radius.

$$\phi = \frac{r_G}{R_e} \quad (5)$$

The Gaussian filter is selected primarily for two reasons: simplicity and comparability. It is widely used [16, 17], which allows for easier interpretation of results and comparison with previous studies.

In Figure 3, the effect of the Gaussian filter is visible when compared to the unfiltered data shown in Figure 2. Applying both filters sequentially significantly reduces noise and processing artifacts, resulting in a much clearer representation of the underlying gravitational signals, as demonstrated in Figure 4. This figure also shows that the most substantial decreases in gravitational acceleration occur near Greenland and Antarctica, consistent with mass loss from ice sheet melting. Additionally, notable increases in gravitational acceleration are observed in the Amazon Basin and the Hudson Bay. The latter exhibits the most pronounced change, attributable to glacial isostatic

adjustment [18].

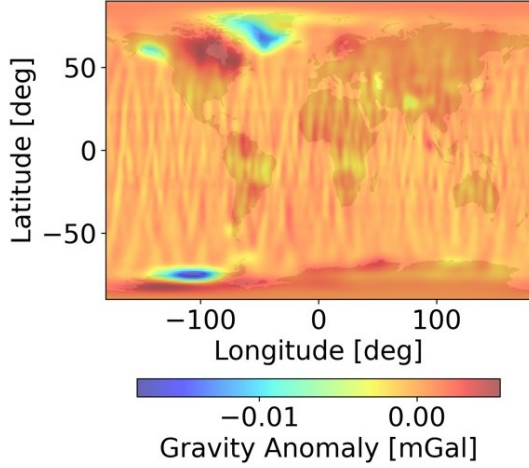


Fig. 3. Filtered change in gravitational acceleration over Earth's surface, shown at 360×180 resolution in milligals (mGal), based on data from April–May 2002 and May–June 2017 only using a Gaussian filter.

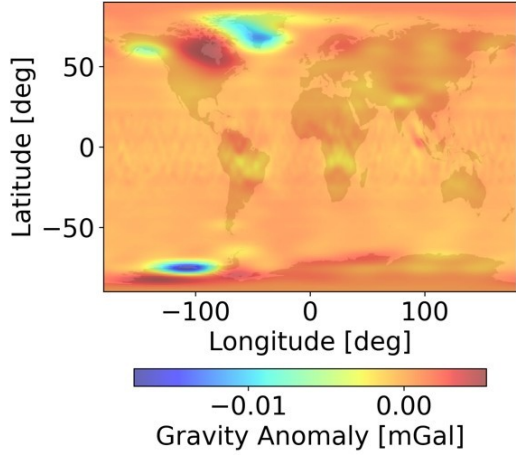


Fig. 4. Filtered change in gravitational acceleration over Earth's surface, shown at 360×180 resolution in milligals (mGal), based on data from April–May 2002 and May–June 2017 using both a Gaussian and correlation filter.

C. Temporal Data Modeling Using Regression Analysis

To capture trends and the spatial distribution of mass changes over Greenland, a linear inverse problem was formulated using Least Squares Regression (LSQR), following the method presented by Crowley et al. [19]. A set of basis functions, representing constant, linear, annual, semiannual, and similar components, was defined, and is displayed in Equation 7. LSQR was then applied to estimate the amplitude of each basis function for every spherical harmonic coefficient.

For each coefficient, an observation vector was formed from its time series. A weighting matrix $\mathbf{W}$ was constructed based on the standard deviations for each monthly solution, as defined in Equation 6. Then, a block matrix $\mathbf{B}$ was assembled by evaluating the basis function vector, shown in Equation 7, at each observation time, as defined in Equation 8. With these matrices defined, the model amplitudes were computed using Equation 9.

$$\mathbf{W} = \begin{bmatrix} 1/\sigma_1^2 & 0 & \cdots & 0 \\ 0 & 1/\sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/\sigma_{\text{end}}^2 \end{bmatrix} \quad (6)$$

$$\mathbf{T} = [1, (t - t_0), \cos 2\pi(t - t_0), \sin 2\pi(t - t_0), \cos 4\pi(t - t_0), \sin 4\pi(t - t_0), \dots] \quad (7)$$

$$\mathbf{B} = \begin{bmatrix} \mathbf{T} & 0 & \cdots & 0 \\ 0 & \mathbf{T} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathbf{T} \end{bmatrix} \quad (8)$$

$$\mathbf{X} = (\mathbf{B}^T \mathbf{W} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{W} \mathbf{Y} \quad (9)$$

The components of the resulting matrix $\mathbf{X}$ represent the least squares solution for the partial coefficients of the single spherical harmonic coefficient. These partial coefficients, when multiplied with their respective basis function and summed up, then equate to the original harmonic coefficient. Therefore, the matrix $\mathbf{X}$ can be visualized as shown in Equation 10, with the different rows representing different spherical harmonics coefficients.

$$\mathbf{X} = \begin{bmatrix} C1_{\text{static}} & C1_{\text{lin}} & C1_{\text{an}_{\cos}} & C1_{\text{an}_{\sin}} & \cdots \\ C2_{\text{static}} & C2_{\text{lin}} & C2_{\text{an}_{\cos}} & C2_{\text{an}_{\sin}} & \cdots \\ C3_{\text{static}} & C3_{\text{lin}} & C3_{\text{an}_{\cos}} & C3_{\text{an}_{\sin}} & \cdots \\ C4_{\text{static}} & C4_{\text{lin}} & C4_{\text{an}_{\cos}} & C4_{\text{an}_{\sin}} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \quad (10)$$

To define the uncertainty of the model, a residual calculation was performed according to method described by De Kleijn [20]. This calculation is displayed in its general form in Equation 11.

$$r_i = y_i - \hat{y}_i \quad (11)$$

Equation 12 describes the residual, as defined in Equation 11, for a specific coordinate input. SH_{val} describes the spherical harmonics coefficients extracted from the Level 2 GRACE data, which provides the actual observed values, while the multiplication of $\mathbf{X} \mathbf{B}$ represents the summation of all of the basis functions with their respective model coefficient. The goal is to minimize these residual values to attain a R^2 value as close as possible to 1.

$$Res(\theta, \lambda) = SH_{val} - XB \quad (12)$$

Using appropriate basis functions, the model typically achieved $R^2 > 0.8$, while unfiltered data produced a maximum $R^2 \approx 0.2$.

D. Equivalent Water Height Area Analysis

The LSQR provides trends in gravitational field coefficients. To estimate mass changes in regions of Greenland, these coefficients are transformed into equivalent water height variations, representing the thickness of a water layer required to produce the observed gravity changes [21].

$$\Delta\sigma(\theta, \lambda) = \frac{R_e \cdot \rho_{avg}}{3} \sum_{l=0}^{\infty} \sum_{m=0}^l \bar{P}_{lm} \cdot \cos(\theta) \quad (13)$$

$$\frac{1 + k_l}{2l + 1} \cdot (\Delta C_{lm} \cos(m\phi) + \Delta S_{lm} \sin(m\phi))$$

Since Equation 13 determines the change in surface density, the spherical harmonic coefficients C_{lm} and S_{lm} in Equation 1 must represent changes in the geoid coefficients, denoted as ΔC_{lm} and ΔS_{lm} , as stated by Wahr et al. [21]. These delta values correspond to the differences between the harmonic coefficients at the reference time $t = 0$ and those at the time point being analyzed, and are computed across the entire considered time span. The normalized Legendre polynomials are then calculated according to Equation 2.

Additionally, the set of Love numbers k_l must be included in the formulation, as they characterize the Earth's response to external, surface, or internal perturbations [22]. In the present context, they account for long-term viscoelastic effects associated with postglacial rebound since the Pleistocene epoch, as well as density variations within the Earth's mantle [23]. The k_l values used correspond exclusively to the elastic response of the Earth and do not incorporate plastic deformation. This approximation is considered acceptable, as plastic deformation only becomes relevant on significantly longer timescales and can thus be neglected [21].

The k_l values applied in the calculations are obtained from Han et al. [23], based on the Preliminary Reference Earth Model [24]. These computations provide Love number values for harmonic degrees up to $l = 200$, which are listed in Table I. Since the values are not available for all degrees, the missing values are estimated through linear interpolation between the known data points. This interpolation introduces errors of less than 0.05% for degrees up to $l = 200$, as noted by Wahr et al. [21].

TABLE I

Love numbers k_l for specific degrees l , computed by Han et al. [23]

l	k_l	l	k_l
0	+0.000	12	-0.064
1	+0.027	15	-0.058
2	-0.303	20	-0.051
3	-0.194	30	-0.040
4	-0.132	40	-0.033
5	-0.104	50	-0.027
6	-0.089	70	-0.020
7	-0.081	100	-0.014
8	-0.076	150	-0.010
9	-0.072	200	-0.007
10	-0.069		

Once a change in density is defined through Equation 13, it can be related to a change in equivalent water height by dividing it by the density of the water, as displayed in Equation 14. Hence, substituting Equation 13 into Equation 14 results in the final formula for the EWH.

$$EWH = \frac{\Delta\sigma}{\rho_{water}} \quad (14)$$

To analyze volume changes over specific regions of the Greenland Ice Sheet, such as individual glaciers, a spatial domain is first defined to represent the area of interest. Within this domain, a meshgrid is constructed at a general resolution of 0.25° , excluding grid points that fall over ocean surfaces to ensure land-only evaluation. For each node within the valid meshgrid, the equivalent water height is computed according to Equation 14.

Subsequently, for each grid cell defined by four adjacent nodes, the mean EWH is calculated. This average value is then multiplied by the surface area of the corresponding cell to estimate the localized volume change in cubic kilometers. Given the approximate equivalence of $1km^3$ of water to 1 gigaton, this value also represents the mass change in gigatons. Summing the contributions from all valid grid cells yields the total volume (or mass) change over the glacier for the specified period.

III. Results

A. Area Determination

To perform an analysis on the mass changes of Greenland in time, its surface area is divided into six subareas. These subareas are roughly defined according to the areas shown in Figure 5. These areas are then analyzed using the area analysis method described in section II. The exact results of this analysis for all the regions, the corner points, and the areas of these zones are tabulated in Table III in the Appendix.

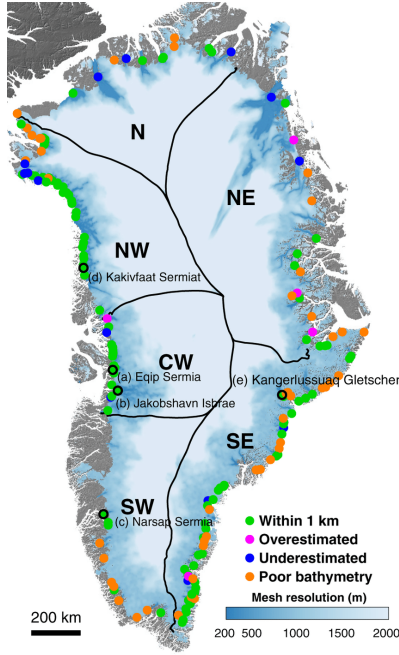


Fig. 5. Map of Greenland partitioned into the six regions used in the analysis. [25]

B. Analysis Results

For all six areas, the analysis described in section II is applied. The volume changes of these areas are evaluated from 2004 to 2017 and are presented in Figures 6 to 11.

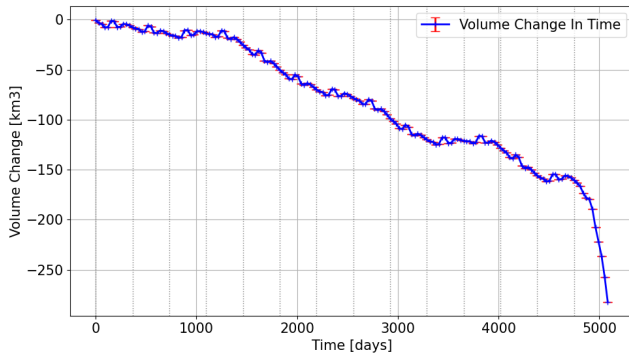


Fig. 6. Evolution of ice mass in the northern reference area of Greenland from 2004 to 2017, shown in km^3 (approximately equal to Gt).

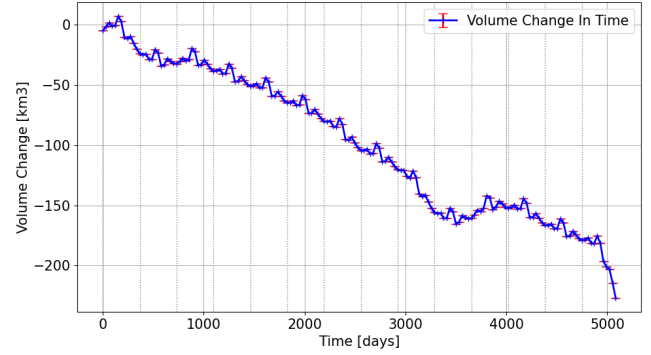


Fig. 7. Evolution of ice mass in the Northeast reference area of Greenland from 2004 to 2017, shown in km^3 (approximately equal to Gt).

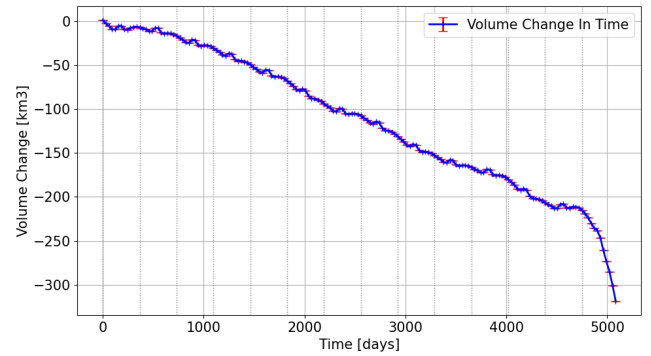


Fig. 8. Evolution of ice mass in the Northwest reference area of Greenland from 2004 to 2017, shown in km^3 (approximately equal to Gt).

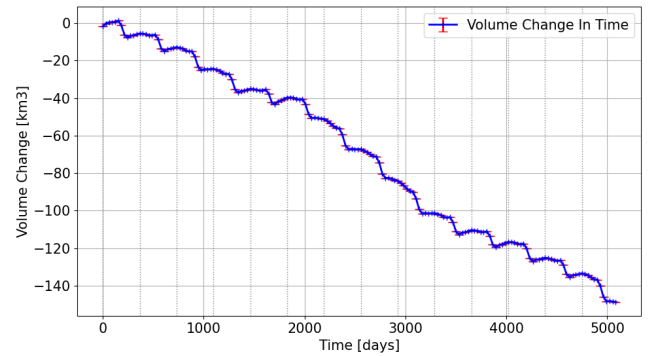


Fig. 9. Evolution of ice mass in the Central West reference area of Greenland from 2004 to 2017, shown in km^3 (approximately equal to Gt).

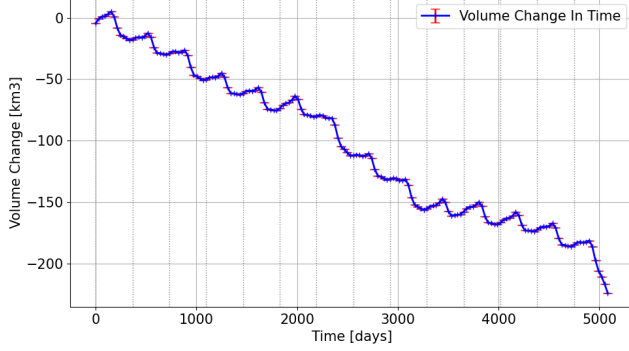


Fig. 10. Evolution of ice mass in the Southeast reference area of Greenland from 2004 to 2017, shown in km^3 (approximately equal to Gt).

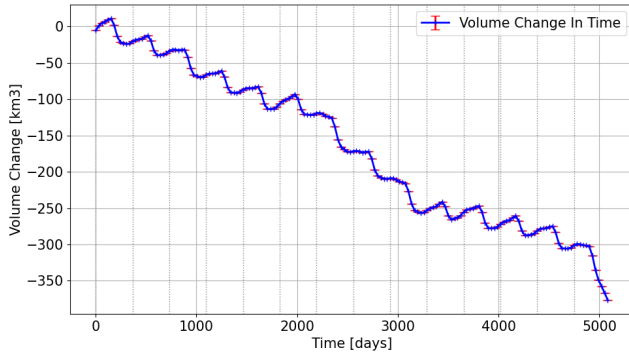


Fig. 11. Evolution of ice mass in the Southwest reference area of Greenland from 2004 to 2017, shown in km^3 (approximately equal to Gt).

In the plots shown in Figures 6 to 11, two clear trends are visible. The most obvious trend is the general downward trend, which describes a clear mass loss during the analyzed period. This trend is visible in all regions and can be explained by global warming, as glaciers and ice caps melt due to increasing temperatures [26]. Another trend is the oscillation in the graphs. This trend corresponds to seasonal changes. During winter months, the ice caps grow, which can be seen by the positive slopes, while during summer months, the ice caps and glaciers melt, which can be seen by the negative slope. The loss during the summer outweighs the gain during the winter, resulting in an overall mass loss, i.e. the first trend. The total volume change between 2004 and 2017 is shown in Table II. It is clear that all considered areas experienced a total volume loss. Given the approximate 1:1 relationship between ice volume and mass, this volume loss can also be interpreted as a mass loss.

TABLE II

Total volume change per analysis region between 2004 and 2017, shown in km^3 ($\approx$ gigatons).

Region	Total volume change [km^3]
North	-285.726
North-East	-226.823
North-West	-318.736
Central-West	-148.770
South-East	-223.969
South-West	-376.422

IV. Discussion

A. Static Gravity Map

In order to validate the equations used to calculate the spherical harmonics, a simple static gravity map can be used. These static gravity maps, when correctly calculated and layered on top of a map of the world, display geological locations of higher and lower gravitational acceleration. Through simple visual analysis of said map, the most significant high-density areas of the globe can be confirmed, for example all big mountain ranges throughout the globe, from the Himalayas, to the Andes, and the Alps. Such a static gravity map is already displayed in subsection II-A, namely as Figure 1. As can be clearly seen in the graph, all these high-density locations are displayed properly and at the correct locations. Additionally, the values of the static gravity field, displayed in $[mGal]$ in Figure 1, also add up to the average gravitational acceleration, since the unit conversion stated in Equation 15 holds and gives values for the static gravity field of the Earth of around 9.80 m/s^2 .

$$1 [mGal] = 1 \cdot 10^{-5} [m/s^2] \quad (15)$$

Based on the reasons outlined above, the results from the spherical harmonics calculations are validated. Furthermore, this confirms that all input values have been computed correctly and can be reliably used for subsequent calculations.

B. LSQR Validation

The next step was to assess the performance of the Least Squares Regression (LSQR) model used to fit the spherical harmonics data. The model accuracy can be defined through the R^2 values for the entire model. The R^2 value has a maximum of 1, which would correlate to a perfect prediction. For the final model, the filtered and unfiltered R^2 values were 0.9319 and 0.3848, respectively. This means that approximately 93 percent of the variance is explained by the model. This implies that the model is accurate, as can also be seen in Figure 12, which shows a comparison between the GRACE data and the linear regression model of the spherical harmonics coefficient $C_{3,2}$.

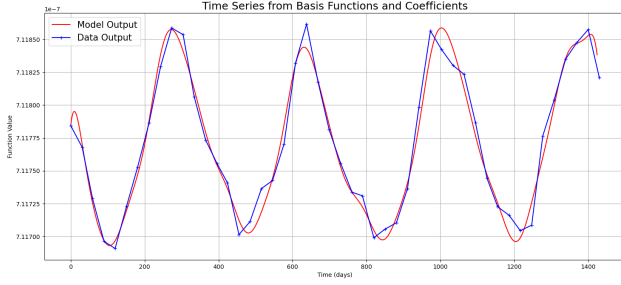


Fig. 12. Comparison of the spherical harmonics coefficient $C_{3,2}$ from GRACE data (blue) with the LSQR model data (red).

C. Ice Sheet Mass Balance Validation

Building on the confirmed accuracy of the LSQR method, the physical outputs of the model are validated by comparison with published mass balance measurements. By analyzing the model results for each region as well as the entire Greenland ice sheet, specific values for mass loss were obtained, as detailed in section III. These results were compared with findings by Mouginot et al. [27], who reported average mass losses of 220 ± 21 Gt/year between 2003 and 2009, and 323 ± 28 Gt/year between 2011 and 2014. Additional data is provided by Velicogna et al. [28], who report an average mass loss of 261 ± 45 Gt/year for the period April 2002 to September 2019. The results of 192 ± 6 Gt/year for the period 2007 to 2012, shown in Figure 13, are just outside these margins, but are within an acceptable order of magnitude, especially since both researches mentioned before have significant error margins, almost 10 percent. Further research could, however, focus on getting a better understanding of the error margin that the mentioned values might have, which could close the gap between the calculated values and the values mentioned above. The difference can also be explained by approximations in area calculations, the size of the area nodes, and consequently the limited representation of complex coastal regions where the mass loss is most significant. Further analysis could focus more on closing that gap even further.

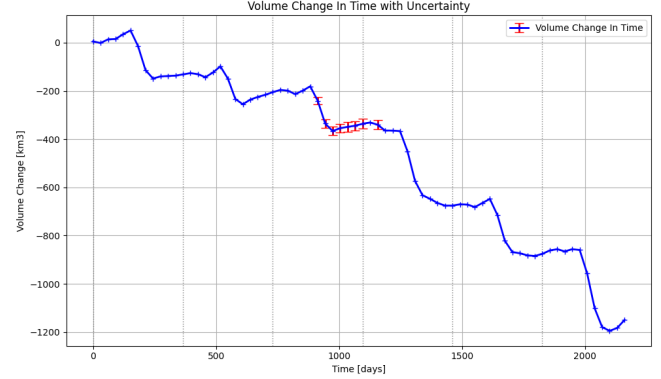


Fig. 13. Evolution of ice volume over Greenland from 2007 to 2012, shown in km^3 (approximately equal to Gt). The blue line indicates the cumulative volume change and the red bars represent the uncertainty range (1σ) at the monthly time steps.

D. Comparison of Results

The graphs in section III clearly show distinct regional differences in ice mass change. For instance, when comparing Figure 8 and Figure 11, the seasonal variations are more pronounced in the South-West region, indicating more dynamic melt and accumulation cycles. This contrast can be attributed to higher average summer temperatures in the South-West, which lead to increased surface melts, while the North remains colder throughout the year, with lower glacial activity and smaller seasonal temperature variations [29]. As a result, the North retains more ice mass annually. This interpretation is supported by the total volume changes shown in Table II, where both regions, despite having comparable surface areas, exhibit significantly different net mass losses.

Another clear pattern in Table II is the greater ice mass loss in the western regions compared to the eastern ones. One of the main reasons for this is the influence of warm Atlantic waters, which reach the western coast via the West Greenland Current [30, 31]. These waters flow into fjords and melt the fronts of marine-terminating glaciers, especially in areas like the South-West, where glaciers such as Jakobshavn Isbræ have shown rapid retreat [32]. Western Greenland also has lower elevations and wider fjords, which make its glaciers more sensitive to ocean heat. In contrast, the eastern side is more mountainous, with narrower fjords that offer some protection from warm water intrusion [33]. These differences in ocean exposure and landscape help explain why ice loss has been more severe in the South-West and North-West, even though the eastern and western regions are located at similar latitudes.

V. Conclusion

The aim of this study was to quantify and compare ice mass loss across distinct regions of Greenland between 2004 and

2017. To achieve this, the level 2 GRACE gravity data was used to derive gravity parameters using spherical harmonics expansions, which were filtered using a low-pass Gaussian filter to reduce the noise. Further, the temporal data was modeled using a least-squares regression analysis to capture trends and spatial distribution of mass changes. Lastly, the trends provided from the analysis were transformed into equivalent water height variations, representing the thickness of water required to cause the gravity perturbations. These volume changes can be interpreted as mass changes due to the approximate 1:1 relationship between ice volume and mass.

As provided in Figures 6 to 11, each distinct region maintains a similar decreasing trend in their respective volume changes. The analysis provides that Greenland lost a total of 1,580 km³ of ice between 2004 and 2017, equating to an equivalent water height of 0.869 m. Furthermore, another matching characteristic between the regions are the oscillations, which correspond to seasonal changes. Although ice mass tends to increase during the winter months, this gain is outweighed by the more substantial melting that occurs during the summer. This trend is evident across all regions and is primarily attributed to global warming [26]. In comparison, the seasonal variations are more pronounced in the South-West region, due to its lower average temperatures, and the West Greenland Current, whereas they are less noticeable in the Northern regions, where seasonal temperature fluctuations are smaller. In particular, the South-West region experienced the greatest ice loss, with 376.42 km³ melting during the study period.

Regarding validation, the aim of this study was well achieved. The spherical harmonics calculations when compared to static gravity maps accurately depicted the high-density areas. Furthermore, the ice sheet mass balance calculated falls within the margins from findings from previous research papers [27, 28]. Additionally, the least squares regression model explained approximately 93 percent of the variance. Thus, this analysis provides valuable and credible information that can support and guide future research. Nevertheless, several extensions remain possible. For example, GRACE gravity data could be used to examine trends in winter growth months (i.e. December, January and February), potentially offering a deeper understanding of seasonal variability and enabling the identification of extreme events. Additionally, extending the study to include specific glaciers across other regions in Greenland, for instance, the Helheim Glacier, one of Greenland's largest glaciers, may offer valuable insights into how glacier size influences changes in ice mass. Such an analysis could contribute to a more comprehensive understanding of the dynamics of ice loss and, in retrospect, the impacts of global warming.

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Appendix

TABLE III

Overview of the analyzed Greenland regions, listing geographic corner coordinates and the considered surface area for each.

	Area [km ²]	Corner points [lon, lat]
North	331737.04	(-72.03, 78.16), (-45.64, 76.39), (-23.16, 82.45), (-25.71, 82.58), (-38.64, 82.65), (-63.66, 81.58)
North-East	573216.8	(-41.08, 72.44), (-41.18, 69.96), (-33.34, 69.86), (-30.79, 69.32), (-23.69, 69.79), (-19.45, 72.94), (-11.82, 81.24), (-21.68, 81.68), (-39.70, 78.43), (-43.20, 76.22)
North-West	260735.55	(-51.68, 71.70), (-37.47, 72.00), (-42.77, 76.15), (-72.25, 78.20), (-70.44, 75.89)
Central-West	109866.77	(-50.30, 68.72), (-42.99, 68.68), (-40.55, 69.92), (-41.08, 72.03), (-47.76, 71.90), (-51.26, 71.43)
South-East	227039.64	(-44.15, 60.58), (-24.01, 69.76), (-25.71, 70.19), (-28.89, 69.72), (-27.61, 70.22), (-35.04, 69.22), (-41.29, 69.62)
South-West	315680.44	(-52.21, 61.65), (-44.26, 60.58), (-41.08, 68.55), (-50.51, 68.55), (-55.50, 65.23)

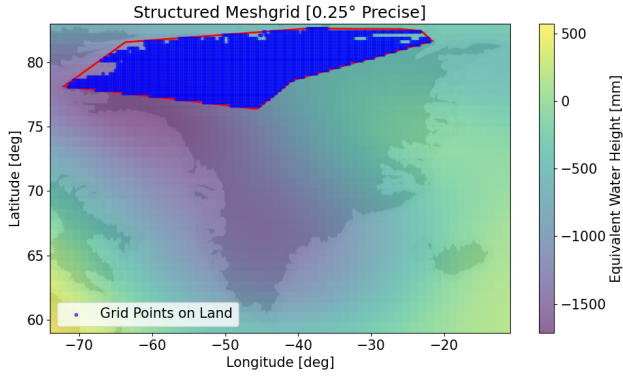


Fig. 14. Zone of interest (red) and corresponding meshgrid (blue) with 0.25° spatial resolution, covering the approximate North Greenland region.

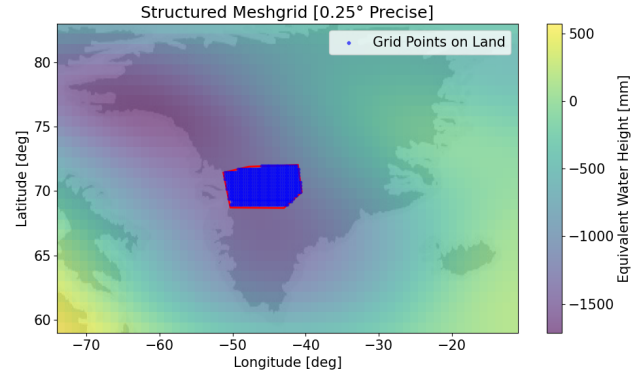


Fig. 17. Zone of interest (red) and corresponding meshgrid (blue) with 0.25° spatial resolution, covering the approximate Central-West Greenland region.

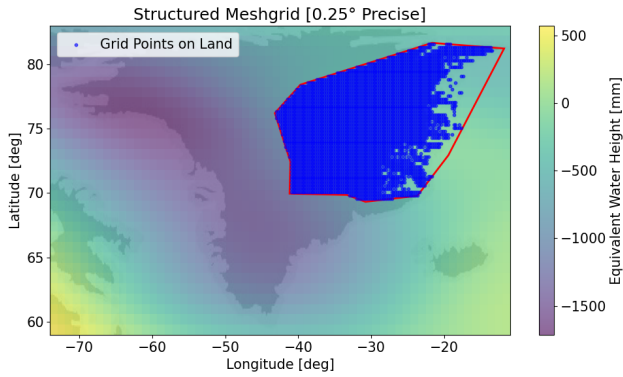


Fig. 15. Zone of interest (red) and corresponding meshgrid (blue) with 0.25° spatial resolution, covering the approximate North-East Greenland region.

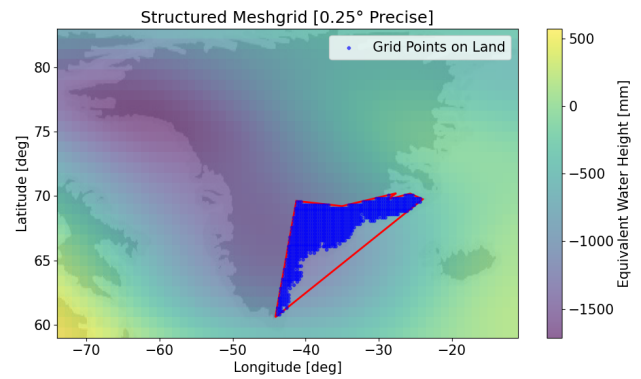


Fig. 18. Zone of interest (red) and corresponding meshgrid (blue) with 0.25° spatial resolution, covering the approximate South-East Greenland region.

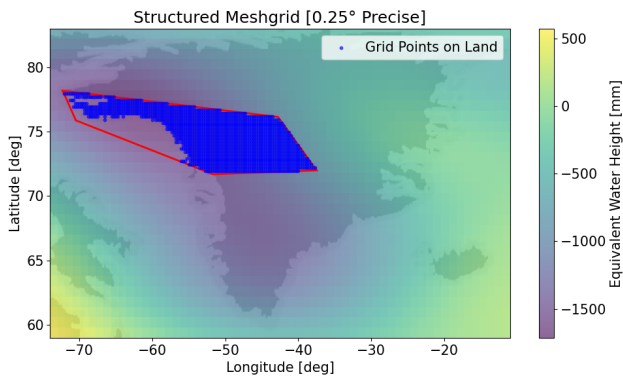


Fig. 16. Zone of interest (red) and corresponding meshgrid (blue) with 0.25° spatial resolution, covering the approximate North-West Greenland region.

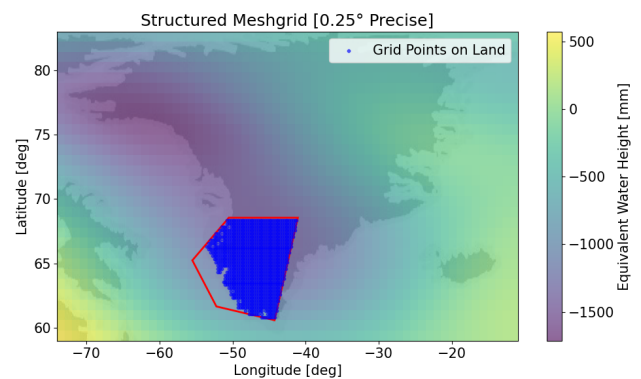


Fig. 19. Zone of interest (red) and corresponding meshgrid (blue) with 0.25° spatial resolution, covering the approximate South-West Greenland region.